# 📄 PRD — Food Delivery & Ordering Platform

## 1. Product Overview

### Product Name

**NamNam**

### Vision

Build a scalable, multi-region food ordering and delivery platform connecting customers, restaurants, delivery agents, and admins, with strong consistency for money and orders, and clear operational ownership.

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## 2. User Types & Roles

### 2.1 End User (Customer)

- Browse restaurants
- View menus
- Place orders
- Choose payment method (Online / Cash on Delivery)
- Track order & delivery status
- View order history

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### 2.2 Restaurant (RBAC REQUIRED)

**Roles inside a restaurant**
**Restaurant Owner**

**Permissions**

- Manage restaurant profile
- Manage menu & pricing
- View all orders
- View financial balance
- View payout history
- Assign / revoke staff roles

**Restaurant Manager Permissions**

- Manage menu
- Accept / reject orders
- Update order status
- View orders & basic analytics

**Restaurant Staff**

**Permissions**

- View incoming orders
- Update preparation status only

RBAC is enforced **per restaurant**, not globally.

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## 2.3 Delivery Agent

- Accept or reject delivery tasks
- Pick up orders from restaurants

- Update delivery status:
    - Assigned
    - Picked up
    - Delivered
- View delivery history
- View earnings (read-only)

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### 2.4 Admin

- Manage restaurants
- Manage delivery agents
- View all orders
- Monitor payments
- Record restaurant payouts

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# 3. Order Lifecycle (End-to-End)

1. Customer places order
2. Payment decision:
    - **Online payment** → payment authorization required
    - **Cash on Delivery (COD)** → no pre-payment
3. Order sent to restaurant
4. Restaurant accepts or rejects

5. Order prepared
6. Delivery agent assigned
7. Order picked up
8. Order delivered
9. Financial settlement recorded

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# 4. Delivery Assignment

## Assignment Model

- **Automatic assignment** (default)
    - Based on proximity
    - Agent availability
- **Manual override** (Restaurant)
    - Reassign agent
    - Handle edge cases

Delivery agents can:

- Accept
- Reject (limited retries)

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# 5. Payments & Money Flow

## Payment Methods

- Online payments

- Cash on Delivery (COD)

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## Online Payment Flow

- Payment authorized before order confirmation
- Idempotent payment handling
- Order created only after confirmed payment

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## COD Flow

- Order created without payment
- Delivery agent collects cash

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## Restaurant Balance Model

- Each restaurant has a **running balance**
- Balance increases when:
    - Order is delivered successfully
- Platform commission deducted automatically

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## Payouts

- Restaurants are paid externally (bank transfer)
- **Admin records payout events in the system**
- Balance reduced after payout is recorded

- Full payout history retained

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# 6. Functional Requirements

## Customer

- Account management
- Restaurant discovery
- Menu browsing
- Cart management
- Order placement
- Payment selection
- Order tracking

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## Restaurant

- Menu CRUD
- Order management
- Status updates
- Financial balance view
- Role management (Owner only)

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### Delivery Agent

- Task assignment
- Status updates
- Delivery history

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### Admin

- Full visibility across system
- Manual overrides
- Financial reconciliation
- Payout recording
- Manage restaurants ● Reports

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# 7. Non-Functional Requirements

- **Availability:** High (order placement & payments are critical)
- **Latency:**
    - Browsing <1s
    - Checkout < 1-3s
- **Consistency:**
    - Strong consistency for orders & payments
    - Eventual consistency acceptable for analytics
- **Scalability:** Must handle peak traffic
- **Security:**

    - Strong authorization boundaries
    - Secure payment handling
- **Multi-Region:**
    - Active traffic in multiple regions
    - Regional reads
    - Strong consistency where required

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# 8. Capacity & Scale (Back-of-the-Envelope)

## Traffic

- **MAU:** 1,000,000
- **DAU:** 200,000
- **Actions / user / day:** 27
- **Requests / day:** ~5.4M
- **Average RPS:** ~62
- **Peak RPS:** ~620

## Read / Write Ratio

- **Reads:** 85%
- **Writes:** 15%

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## Storage

- **Orders growth:** ~146 GB / year

### Bandwidth

- **~270 GB / day**

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# 9. Data Retention

- Orders: retained long-term, we show in the customer app and restaurant only current year orders. Others are stored in another database storage
- Payments, same as orders
- Logs: time-limited (30 days)
- Analytics data: aggregated & archived

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# 10. Multi-Region Assumptions

- Users served from nearest region
- Restaurants tied to a primary region
- Orders processed in restaurant's region
- Financial data requires strong consistency

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# 11. Constraints & Risks

- Meal-time traffic spikes
- Payment provider failures
- Delivery agent shortages

- Cross-region consistency complexity
- Settlement reconciliation errors

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## 12. Success Criteria

- Orders complete end-to-end without loss
- Payments are correct and auditable
- Restaurant balances are accurate
- System remains stable under peak load
- Clear ownership and authorization boundaries

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## 13. Out of scope

- Recommendation systems
- Loyalty programs
- AI-based delivery optimization
- Reviews